EXPERIMENT - 3

Aim: To perform the user's view analysis for <case study name>: Use case diagram.

Theory:

1) What is Scenario based modeling?
2) How to list down the use cases and Actors in a use case diagram
3) Guidelines for finding out the relationship between Use cases.

Scenario-Based Modeling

- Makes use of use cases to capture the ways in which end-users will interact with the system
- UML requirements modeling begins with the creation of scenarios in the from of use cases, activity diagrams, and swim lane diagrams

Developing Use Cases

- Use cases capture the interactions between actors (i.e. entities that consume or produce information)
- Begin by listing the activities performed by a single actor to accomplish a single function
- Continue this process for each actor and each system function
- Use-cases are written first in narrative form and then mapped to a template if more formality is required
- Each primary scenarios should be reviewed and refined to see if alternative interactions are possible
    - Can the actor take some other action at this point?
    - Is it possible that the actor will encounter an error condition at some point? If so, what?
    - Is it possible that the actor will encounter some other behavior at some point? If so, what?

Exceptions:

- Describe situations (failures or user choices) that cause the system to exhibit unusual behavior
- Brainstorming should be used to derive a reasonably complete set of exceptions for each use case
    - Are there cases where a validation function occurs for the use case?
    - Are there cases where a supporting function (actor) fails to respond appropriately?
    - Can poor system performance result in unexpected or improper use actions?
- Handling exceptions may require the creation of additional use cases

Procedure:

1) Analyze the system to find out the various functionalities.
2) List down the set of Use Cases and the various Actors in the system to be developed.
3) Draw a Use Case diagram and draw the various associations between them.

Part B:

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Actor	**Classification**	**Role Description & Goal**
Patient	Primary Actor	Logs daily vitals, manages reproductive health tracking, tracks prescriptions, and uploads medical reports.
Doctor / Dietician	Primary Actor	Remotely monitors patient vitals, reviews diet and lab records, and adjusts medical treatment plans.
Caretaker / Nurse	Primary Actor	Manages medication dosages/schedules, conducts check-ins, and responds to critical vital anomalies.
Patient's Family	Primary Actor	Supervises caregiver check-in logs and receives emergency high-priority health notifications.

Functional requirements from user stories and project documentation map directly to system use cases:

- UC1: User Authentication & Role Verification (RBAC access login)
- UC2: Manage Access Permissions & Consent (Granting doctor/caregiver viewing rights)

- UC3: Log Daily Vitals (Blood pressure, heart rate, blood glucose, temperature)
- UC4: Track Menstrual & Reproductive Health (Symptom logging, cycle predictions)
- UC5: View Medication Schedule & Reminders (Daily prescription checklist)
- UC6: Upload Documents to Secure Vault (Encrypted storage of lab reports/prescriptions)
- UC7: View Patient Health & Diet Dashboard (Real-time vitals and progress charts)
- UC8: Adjust Treatment & Diet Plans (Updating clinical recommendations)
- UC9: Download & Review Lab Reports (Accessing encrypted medical history)
- UC10: Update Medication Dosage & Schedule (Modifying dosage timings and amounts)
- UC11: Log Caregiver Check-Ins & Notes (Recording visit notes and updates)
- UC12: View Caregiver Oversight & Activity Logs (Chronological check-in history)
- UC13: Receive Anomaly & Missed Medication Alerts (High-priority alert processing)
- UC14: Validate Input Data (Ensuring numeric and boundary integrity)
- UC15: Trigger Emergency Anomaly Alert (Automated threshold notification dispatch)
- UC16: Encrypt Document (Securing sensitive cloud-stored files)

Actor	**Associated Use Case**
Patient	UC1, UC2, UC3, UC4, UC5, UC6
Doctor/Dietician	UC1, UC7, UC8, UC9
Caretaker/ Nurse	UC1, UC7, UC10, UC11, UC13
Patient’s Family	UC1, UC12, UC13

Output: A Use case diagram for the System.

